Abhishek Vinayak Pai

avpai4@illinois.edu | abhipai.github.io | (217)-904-2753

RESEARCH INTERESTS

At the intersection of **cyber-physical systems**, **learning-based control** and **formal methods**, for safe autonomy in complex environments. Also interested in adversary detection, constrained learning-based planners for multi-agent systems.

EDUCATION

University of Illinois Urbana-Champaign — Aug. 2024 – May 2026 (Expected)
*MS Aerospace Engineering, GPA 3.47/4, **Advisor**: Dr. Sayan Mitra (Dept. of ECE)* — Urbana, IL
***Thesis**: Formally verifiable vision based-controllers for quadrotors*

Manipal Institute of Technology, Manipal — Aug. 2020 – Jul. 2024
B.Tech Mechanical Engineering, Minor in Computational Mathematics, GPA 8.1/10 — Manipal, India

RESEARCH EXPERIENCE

Coordinated Science Laboratory — Aug. 2025 – Present
Graduate Research Assistant — Urbana, IL

- Developing formally verifiable visual-stabilization controllers for drones that provide formal stability guarantees despite bounded visual estimation errors.
- Building the core autonomy stack for Illinois' first high-speed autonomous drone racing platform, including vision-based state estimation and controllers for agile flight.

MAHE Centre for Aerospace Science and Technology (CAST) — Dec. 2023 – Jun. 2024
Research Assistant, Undergraduate Thesis — Manipal, India

- Developed numerical optimization algorithms for hybrid airship shape generation; used CFD to generate and analyze aerodynamic properties. Published in *The Aeronautical Journal* .

SELECTED PROJECTS []

Safe RL for Autonomous Drone Racing — Oct. 2025 – Dec. 2025
University of Illinois Urbana-Champaign, Course Project — Urbana, IL

- Developed an end-to-end time-optimal RL framework for autonomous drone racing using PPO with CBF–based reward shaping for collision-free gate traversal while enabling aggressive racing lines via dynamic safety funnel constraints.
- Achieved ~38% reduction in lap time and ~34% improvement in target-region accuracy compared to the baseline.

Developing and Evaluating VLA+RL Models for Robotic Control — Jan. 2025 – May 2025
University of Illinois Urbana-Champaign, Course Project — Urbana, IL

- Applied concept probes and causal interventions to investigate failure modes in VLA models fine-tuned with RL.
- Achieved ∼81% nominal success on physical manipulation tasks (SO-100), while exposing emergent adversarial strategies with a 92% detection rate.

Autonomous SLAM and Object Detection in Unknown Environments — Mar. 2025 – May 2025
University of Illinois Urbana-Champaign, Course Project — Urbana, IL

- Developed custom LiDAR SLAM with intensity filter to reduce map noise by ∼40% and achieve sub-10cm wall precision.
- Integrated real-time object detection with map-based location reporting for precise object localization.

SELECTED RESEARCH []

Safe Landing Zone Detection for UAVs using Image Segmentation and Super Resolution — **MVA 2023**
Anagh Benjwal, Aditya Vadduri*, Prajwal Uday*, **Abhishek Pai**** — ***[Cited: 1]*** Hamamatsu, Japan

- Novel approach for safe landing zone detection for UAVs, leveraging DL-based image segmentation paired with SuperResolution, resulting in up to 6.3% improvement in accuracy over SOTA methods.

Precise Payload Delivery via UAVs using Object Detection Algorithms ***[Cited: 4]*** — **AICECS 2023**
*Aditya Vadduri, Anagh Benjwal, **Abhishek Pai**, Elkan Quadros, Prajwal Uday* — Manipal, India

- Novel navigation method that improves the accuracy of conventional navigation methods by incorporating a DL-based computer vision approach, achieving a 5x increase in average horizontal precision over conventional GPS-based approaches.

WORK EXPERIENCE

Skyrobee Aerolabs LLP Apr. 2023 – Jun. 2024
Co-Founder & Hardware Engineer Pune, India

- Led the R&D and systems integration of multi-rotor UAVs, developing a real-time navigation stack for autonomous security missions. Secured multiple defense contracts for FY 2023-24.

Airbus ODC at Quest Global Jun. 2023 – Jul. 2023
Software Intern Bangalore, India

- Developed an automated composite-structure defect evaluation software tool, reducing engineering analysis effort by 70%, supporting concession approvals.

LEADERSHIP & MENTORSHIP

Teaching Assistant Jan. 2026 - Present
Dept. of Electrical & Computer Engineering, UIUC Urbana, IL

- Contributing to the revamp of ECE 484: Principles of Safe Autonomy by integrating sota methods in state estimation and perception–control, and introducing a new quadrotor platform for experiential learning.

Vice President Sept. 2025 - Present
Illinois Aerial Robotics, RSO Urbana, IL

- Leading the university's official team for the A2RL drone racing competition, directing technical strategy and operations.
- Co-founded the RSO and raised $2,000 in initial funding to finance hardware, operational and logistical costs.

Autonomous Vehicle Design Head Jan. 2022 - Aug. 2023
AeroMIT, RSO Manipal, India

- Led the design and development of over 20 fixed-wing and multirotor UAVs across diverse missions. Mentored a subsystem team of 8 and contributed to award-winning international competition entries.

Workshop Instructor Oct. 2023
IE Aerospace × AeroMIT, MIT Manipal Manipal, India

- Instructed a workshop on UAV design and aerodynamic analysis, engaging 50+ students across multiple departments.

ACHIEVEMENTS & SCHOLARSHIPS

- Placed **2nd** in design, **4th** overall globally at SAE-Lockheed Martin Aerodesign 2023, Fort Worth, TX.
- Placed **1st** in design, **2nd** in mission performance & **2nd** overall globally at SAE-Lockheed Martin Aerodesign 2022, CA.
- Placed **1st** overall nationwide at the SAE India Aerothon 2022, Bangalore, India.
- Awarded the MAHE *Undergraduate Research Seed Grant* of $2500 to support independent project development and prototyping for drone projects.
- Awarded the *Manipal Merit Scholarship* in 2020 for undergraduate studies.

SKILLS

Programming & Tools: Python, C/C++, MATLAB
Libraries & Frameworks: ROS/2, PyTorch, JAX, OpenCV, PX4, MAVROS
Simulation: Gazebo, Pybullet, MuJoCo
Hardware: NVIDIA Jetson, VOXL2, Raspberry Pi, Pixhawk, SO-100/1

KEY COURSES UNDERTAKEN

Robotics: Formal Methods in Aerospace Robotics, Reinforcement Learning, Estimation of Dynamical Systems, Principles of Safe Autonomy, Computational Systems Engineering

Math & Statistics: Statistical Learning, Time Series Analysis, Computational Linear Algebra, Probability & Design of Experiments, Metaheuristic & Multi-Objective Optimization

OTHER INTERESTS

- Skilled at RC flying of both fixed wing UAVs and multi-rotors.
- Indian classical vocalist and multi-instrumentalist with over 8 years of stage experience. Cleared 3 levels of ABGMV vocals examination with distinction.